Problem 6.1 — Picking-Time Neural Network

Engineering Artificial Intelligence · Exercise Report

1. Setup and Data

The dataset contains 400 synthetic warehouse picking-time observations with 5 features (distance, load, congestion, battery, aisle_width). The true relationship includes three nonlinear effects: a congestion×distance interaction term, a sharp 12-second battery-threshold penalty at 20%, and a mild quadratic term in distance. These make the task impossible to solve well with a linear model alone. 20% of examples (80) are held out as a final test set that is never seen during training or model selection. The remaining 320 examples form the cross-validation pool.

2. Network Architecture and Training

A feedforward network is defined with three hidden layers: **Input(5)** → **32 (ReLU)** → **16 (ReLU)** → **8 (ReLU)** → **Output(1)**. The output layer has no activation (regression task). Weights are initialised with He initialisation (PyTorch default for nn.Linear). Training uses the Adam optimiser (lr = 10^{-3}) and MSE loss for 200 epochs per fold. Features and targets are normalised to zero mean and unit variance using *training-fold statistics only* to prevent data leakage into the validation fold.

3. Five-Fold Cross-Validation Loss Curves

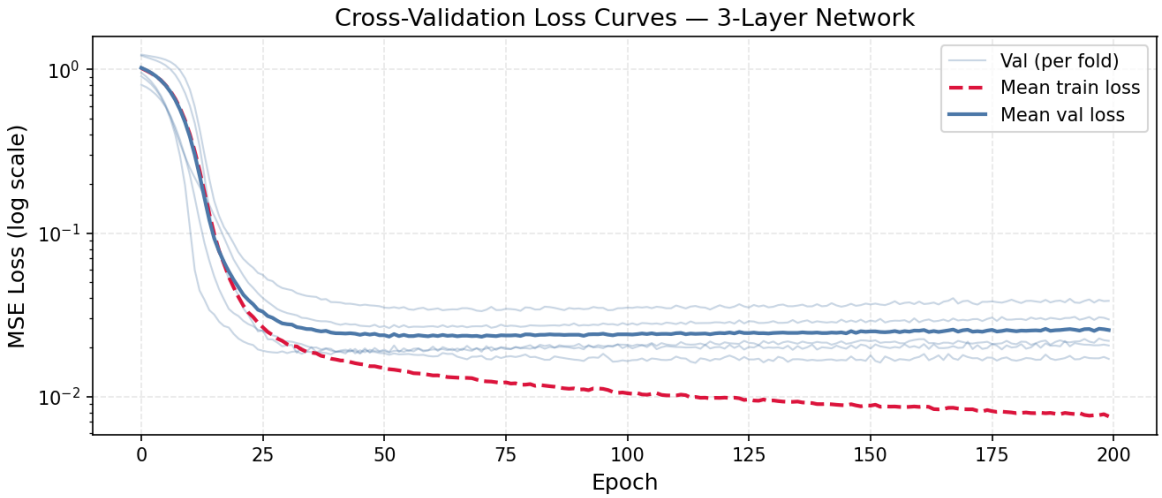

Figure 1. Per-fold validation loss curves (thin blue) and mean training / validation curves (dashed red / solid blue) on a log scale. CV RMSE: 2.64 ± 0.37 sec.

The validation loss decreases steeply in the first ~50 epochs and flattens thereafter. The training loss continues to drop slightly below the validation curve — a mild overfitting signature. The gap is modest relative to the fold-to-fold variance, indicating the chosen architecture is well-matched to the dataset size of 320 training examples. No early stopping was needed at this capacity level.

4. Predicted vs. Actual Plots

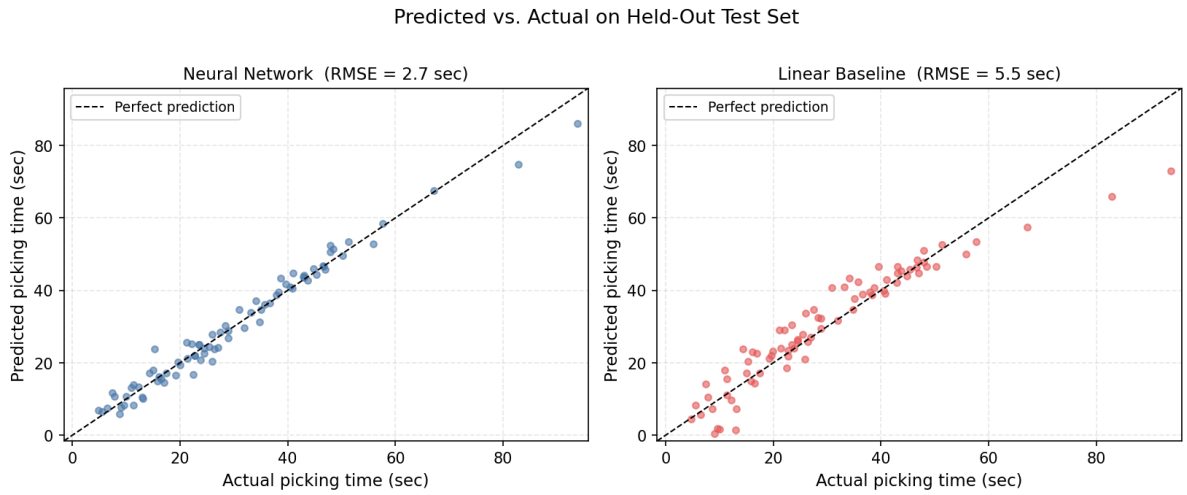

Figure 2. Predicted vs. actual picking time on the 80-example held-out test set for the neural network (left) and linear baseline (right). Points near the dashed diagonal indicate accurate predictions.

The neural network predictions cluster tightly around the diagonal across the full range of picking times. The linear model shows two systematic failure modes: (i) it under-predicts high-congestion / far-distance examples where the interaction term 0.4×congestion×distance dominates, and (ii) it misses the 12-second battery-threshold step entirely, producing a smooth slope in battery instead of the sharp discontinuity at 20%.

5. Linear vs. Neural Network — RMSE Comparison

Model	Test RMSE (sec)	MSE Improvement
Linear baseline (OLS)	5.45	—
Neural network (3 layers)	2.72	75.1 %

The neural network reduces test RMSE from 5.45 sec to 2.72 sec — a 75% reduction in MSE. The remaining error (~2.7 sec) is close to the noise floor of the data (observation noise standard deviation $\sigma = 2$ sec), indicating the network has learned the true relationship well.

6. Depth Experiment

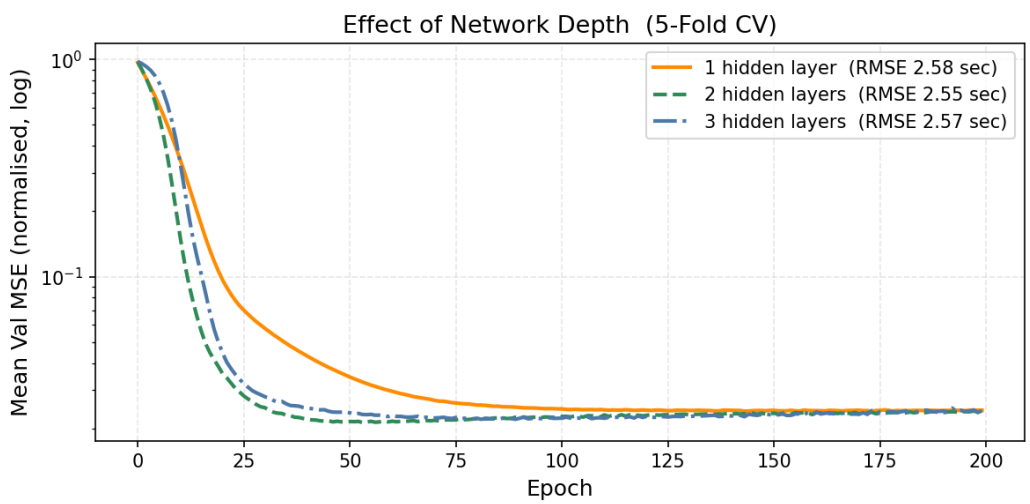

Figure 3. Mean validation loss curves (5-fold CV) for networks with 1, 2, and 3 hidden layers. All curves converge to similar final values.

Hidden layers	Architecture	CV RMSE (sec)
1	5 → 32 → 1	2.58 ± 0.32
2	5 → 32 → 16 → 1	2.55 ± 0.30
3	5 → 32 → 16 → 8 → 1	2.57 ± 0.32

All three depths achieve essentially the same CV RMSE (~2.55–2.58 sec), well within one standard deviation of each other. This finding is consistent with the universal approximation theorem: even a single hidden layer with sufficient width can approximate the target function. The nonlinearities present in this dataset (an interaction term, a threshold function, and a quadratic) are all relatively simple and do not require the hierarchical feature extraction that makes depth beneficial for image or sequence data. Adding layers did not overfit because the total parameter count remains modest (the deepest network has fewer than 700 parameters) relative to 320 training examples.

Final test RMSE — neural network: **2.72 sec** | linear baseline: **5.45 sec** | noise floor: ~2 sec